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B

n is the number of turns per unit length of solenoid.

$$n = \frac{N}{\text{solenoid length}} = \frac{4000}{200 \times 10^{-3}} = 20000 \text{ turns per unit length}$$

$$B = \mu_0 n I = \mu_0 n \left(\frac{V}{R} \right)$$

$$= (4\pi \times 10^{-7})(20000) \frac{6.0}{(3.26 \times 10^3)}$$

$$= 4.6 \times 10^{-5} \text{ T}$$

Option A: uses $R = 3.26 \Omega$ instead of $R = 3.26 \text{ k}\Omega$

Option C: wrong n , use $n = 4000$ turns and wrong $R = 3.26 \Omega$ instead of $R = 3.26 \text{ k}\Omega$

Option D: wrong n , use $n = 4000$ turns.

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D

By Faraday's Law